Huawei Access Node Optical Transport Failure Resolution Procedure for MG Road Core

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_3756

Description

The MG-ACC-46 node, situated at the MG Road Core site, operates as an access node within the telecommunications network, facilitating the convergence of various services onto the optical transport infrastructure. The Optical Transport Failure (ALARM_3756) in this context typically arises from SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which can severely impede service delivery. This failure domain is intricately linked to the physical and optical layers of the network, where the Huawei equipment utilizes specific protocols and hardware mechanics to ensure data transmission integrity. The Huawei CLI dialect is pivotal in diagnosing and resolving these issues, given its vendor-specific commands that interact with the device's control and data planes. Understanding the optical transport layer's behavior, including signal attenuation, dispersion, and the impact of external factors like temperature and humidity, is crucial for effective troubleshooting. The interplay between the optical and electrical components, such as the laser drivers, photodetectors, and the clock and data recovery mechanisms, must also be considered. Given the critical role of the access node in service provisioning, prompt and accurate diagnosis and resolution of optical transport failures are essential to minimize service downtime and ensure the quality of experience for end-users.

Pre-requisite Checks

- Dispatch remote hands to MG Road Core for on-site inspection and intervention if necessary.

- Verify the status of redundant links to ensure no single point of failure exists.

- Check environmental conditions such as temperature and humidity to rule out external factors contributing to the failure.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Loss Due to Misconfiguration or Hardware Failure	Low

Diagnostic Commands & Expected Outputs

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Command: display interface, Expected Output: Interface status, including the current state and any error messages., Description: This command provides an overview of the interface status, helping to identify any interfaces that are down or experiencing errors, which can indicate the source of the optical transport failure.
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Command: display optical-parameters, Expected Output: Optical power levels and other parameters for the optical interfaces., Description: By examining the optical parameters, technicians can assess whether the signal strength is within acceptable ranges and identify potential issues with the optical transmission, such as low power levels or high error rates.

Command: display alarm all, Expected Output: A list of all current alarms on the device, including their severity and description., Description: Reviewing all current alarms can help in understanding the scope of the issue and identifying any related problems that may not be immediately apparent from the primary alarm.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP Module, Command: replace sfp/xfp module , Description: If the diagnostic process reveals that the SFP/XFP module is degraded, replacing it with a new module can resolve the optical transport failure. This step requires careful handling to avoid damaging the module or the device.

Step 2: Step: Repair or Replace Damaged Fiber, Command: N/A, Description: For fiber cuts or damage, the affected section of the fiber needs to be repaired or replaced. This may involve splicing the fiber or running a new fiber segment, and it typically requires specialized equipment and expertise.

Step 3: Step: Adjust Rx/Tx Power Settings, Command: set optical-power , Description: If the issue is due to misconfigured Rx/Tx power settings, adjusting these settings to the recommended levels can resolve the problem. The specific command and acceptable power levels will depend on the device model and the optical link characteristics.

Post-Check Verification

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Command: display interface, Expected Output: Updated interface status showing that the previously faulty interface is now operational., Description: After completing the resolution steps, re-running the interface display command can confirm whether the optical transport failure has been successfully resolved.

Command: display optical-parameters, Expected Output: Updated optical parameters indicating healthy signal levels and minimal errors., Description: Verifying the optical parameters post-resolution ensures that the optical transmission is stable and within specifications, providing a clear indication of the fix's effectiveness.
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Roll-back Procedures

- Step: Revert to Original SFP/XFP Module, Command: replace sfp/xfp module , Description: If the replacement module does not resolve the issue or introduces new problems, reverting to the original module may be necessary to restore the previous state while alternative solutions are explored.

- Step: Restore Original Fiber Configuration, Command: N/A, Description: In cases where fiber repair or replacement was attempted but did not yield the desired results, restoring the original fiber configuration may involve re-splicing or re-routing the fiber to its previous state, which can be complex and time-consuming.

- Step: Revert Rx/Tx Power Settings, Command: set optical-power , Description: If adjusting the Rx/Tx power settings did not resolve the issue, reverting these settings to their previous values can help in isolating the problem further or preparing the system for an alternative fix.